

Worth its weight in carbon? An Analysis of Net-Positive Machine Learning in Grid Carbon Forecasting

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Background

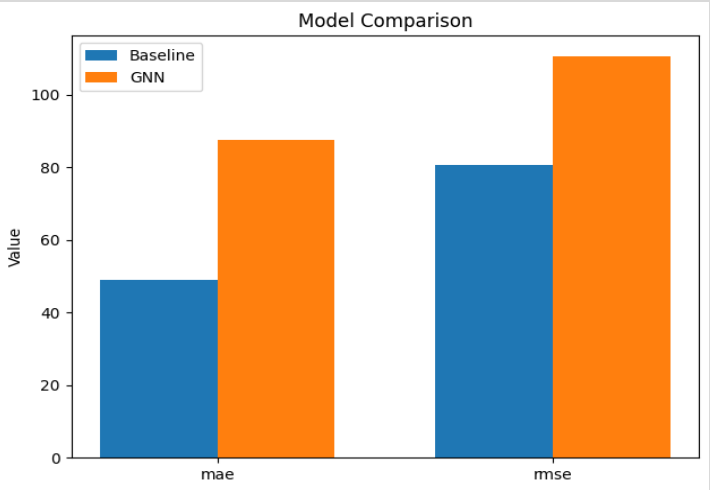
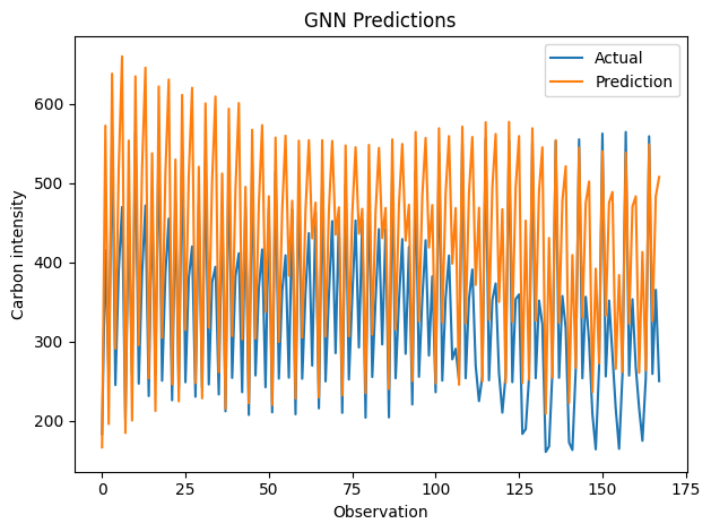
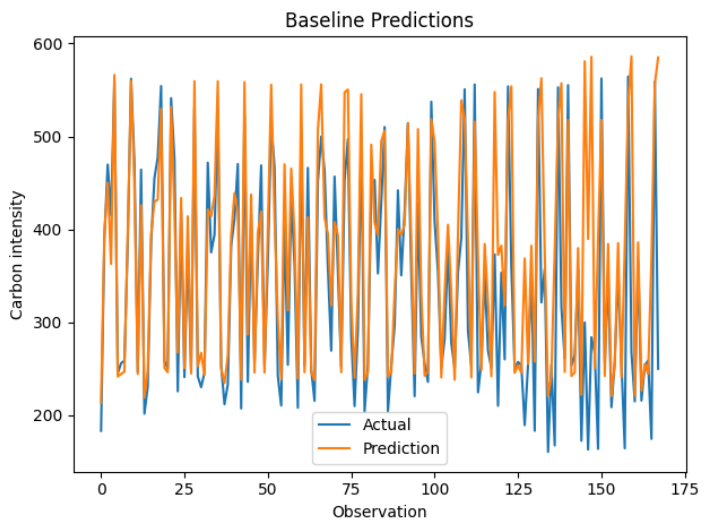
Machine learning is increasingly used in climate and energy applications, including electricity forecasting, grid optimization, and demand shifting. However, training and running these models also consumes electricity and produces emissions, creating a tension between the environmental benefits AI may enable and the carbon cost of computation itself.

Research Question

Can a machine learning model for day-ahead grid carbon intensity forecasting reduce emissions enough to justify its own computational carbon footprint? More specifically, does a graph neural network provide enough value over a simpler baseline model to make its added complexity environmentally worthwhile?

Data

This project uses hourly electricity generation by fuel from the U.S. Energy Information Administration (EIA) Grid Monitor API for major ISO/RTO regions in the continental United States. Fuel-specific emissions factors and regional geographic reference data come from EPA eGRID. Computational emissions from model training are tracked with CodeCarbon.



Model	MAE	RSME	Training Emissions (mg)	Energy Consumed (Wh)	Sessions to break Even
baseline	48.997	80.686	6.3	0.006	0.000043
gnn	87.643	110.634	627.4	0.265	0.000975

Methods

Hourly carbon intensity was estimated by combining regional electricity generation mix with fuel-specific emissions factors. A model-ready dataset was created with time features, lag features, and a day-ahead target, then used to train two models: an XGBoost baseline and a graph neural network. The GNN incorporated approximate spatial structure by mapping EIA regions to adjacent eGRID subregions. Models were then evaluated both on forecast error and on a simple EV-charging shift simulation that estimated emissions savings from charging at the predicted lowest-carbon hour.

Results

The baseline gradient boosting model outperformed the GNN on forecasting accuracy, with **MAE = 48.997** and **RMSE = 80.687** compared with the GNN's **MAE = 87.644** and **RMSE = 110.634**. The baseline also had a much smaller training footprint, at about **6.3 mg CO2eq**, while the GNN emitted about **627 mg CO2eq** during training. The charging simulation showed that both models could identify lower-carbon charging windows in some cases, but the overall forecasting and carbon-cost results still favored the baseline. The current results suggest that the simpler baseline model delivers the better balance of predictive performance and computational carbon cost.

Limitations

This analysis uses a simplified regional graph structure and an approximate mapping between EIA regions and eGRID subregions. The EV-charging simulation is intentionally simple and is based on a limited evaluation window, so estimated emissions savings should be interpreted cautiously. Future work could use a longer time period, stronger interconnection data, and a more realistic deployment scenario for operational decision-making.